LOKSABHA GENERAL ELECTIONS AT CANDIDATE LEVEL

1. Project Overview and Objective

This project involves end-to-end data engineering and analytics on a comprehensive, candidate-level raw dataset of Indian Lok Sabha General Elections Utilizing Excel for robust data cleaning, structural transformation, and pre-processing, the refined data was integrated into Power BI to build a dynamic, multi-page interactive dashboard. The final model translates complex electoral datasets into clear, digestible visual insights detailing political trends and candidate performances.

The primary objective is to demonstrate advanced data pre-processing workflows and data modelling techniques using Excel, coupled with an executive-level Power BI dashboard deployment. By engineering key political metrics such as candidate vote distributions, party seat-to-vote shares, and regional turnout variations the dashboard aims to uncover underlying voting patterns, party efficiencies, and competitive margins to drive data-informed electoral analysis.

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2. Data Sources

- **Source Description and Timeline:** indiadataportal.com/p/elections-in-india/r/eci-general_election_results-pc-qq-vnq **and** 2009-2019.
- **Domain**: Election Commission of India

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3. Problem Statement

- To analyse electoral performance across states, constituencies, and political parties to identify emerging voting trends and seat-sharing efficiencies.
- To study candidate-level data patterns across different election types (Lok Sabha and By-elections) and years to map historical political shifts and regional dominance.
- To evaluate voter participation and turnout factors across diverse constituencies to uncover patterns in democratic engagement and victory margins.
- To understand voter preferences and mandate metrics (such as vote share percentages versus actual seats won) to enhance post-election analytics and predictive modelling strategies

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Election Year	Numeric (Integer)	Year in which the election was conducted

Election Category	Categorical	Type of election (General Election / By-Election)
State Name	Categorical	Name of the State or Union Territory where the election was held
Constituency Name	String (Text)	Name of the parliamentary constituency
Constituency Type	Categorical	Category of constituency (General, SC, ST, etc.)
Candidate Name	String (Text)	Name of the contesting candidate
Party Code	Categorical	Political party abbreviation/code of the candidate
Candidate Votes	Numeric (Integer)	Total number of valid votes secured by the candidate
Candidate Position	Numeric (Integer)	Final rank secured by the candidate in the constituency (1 = Winner)
Total Electors	Numeric (Integer)	Total number of registered voters in the constituency
Total Votes Polled	Numeric (Integer)	Total votes cast in the constituency
Turnout Percentage	Numeric (Decimal)	Percentage of voters who participated in the election
Vote Share (%)	Numeric (Decimal)	Percentage of votes received by the candidate out of total valid votes
Victory Margin	Numeric (Integer)	Difference in votes between the winner and runner-up
Winning Status	Categorical	Indicates whether the candidate won or lost the election
Gender	Categorical	Gender of the candidate (Male/Female/Other)

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5. Tools & Technologies

- **Excel:** Data cleaning, transformation, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights.
- **Fact and Dimension Table:** Converted data into main fact and dimension tables

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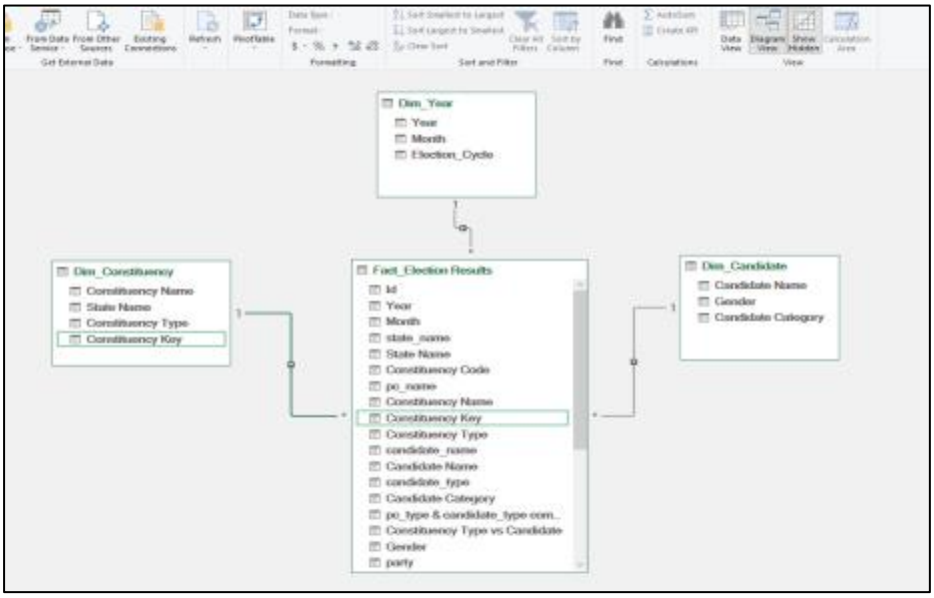
7. Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary using Star Schema Method



- **Calculated Columns & DAX Measures:** Developed advanced DAX measures to calculate election KPIs such as Total Elections, Total Candidates, Total Votes Polled, Average Turnout %, Party Vote Share, Party Seat Share, Seats Won, Winning Rate %, Winners Count, and Average Victory Margin. These calculations facilitated in-depth analysis of electoral trends, party performance, voter participation, and candidate success across Lok Sabha General Elections and By-Elections.

List of DAX measures Used:

Measure Name	Purpose
Total Elections	Count of distinct election years
Total Candidates	Total number of contesting candidates
Total Constituencies	Count of unique constituencies
Total Votes Polled	Total valid votes cast across elections
Average Turnout %	Average voter participation percentage
Total Parties	Count of unique political parties
Winners Count	Total number of winning candidates
Average Victory Margin	Average vote difference between winner and runner-up
Candidates Contested	Number of candidates fielded by a party
Seats Won	Number of constituencies won by a party
Winning Rate %	Seats Won ÷ Candidates Contested
Party Vote Share %	Party Votes ÷ Total Votes
Party Seat Share %	Seats Won ÷ Total Seats Won
Vote Share %	Candidate Votes ÷ Total Constituency Votes
Candidate Count	Total candidates for rank distribution analysis

8. Analysis and Visualizations (Power BI)

Dashboard Features:

- Created an interactive Power BI dashboard using multiple visualizations including KPI Cards, Bar Charts, Column Charts, Donut Charts, Line Charts, Maps, Gauge Charts, Matrix, and Tables to present election insights effectively.
- Implemented Slicers, Filters, and Key Influencers to enable dynamic and user-friendly exploration of election data.

9. Insights & Conclusions

- **Key Findings:**

The analysis of Lok Sabha General Elections revealed significant patterns in voter participation, candidate performance, party dominance, seat distribution, and regional voting behaviour across different election years. The dashboard highlights variations in voter turnout, electoral competitiveness, party-wise vote shares, and constituency-level election outcomes, enabling a comprehensive understanding of India's electoral landscape.

- **Analysis insights:**
  - **Descriptive**

  Identified the total number of elections, candidates, constituencies, political parties, and votes polled across the analysed period. Examined voter turnout trends over different election years.

  - **Diagnosis**

  Investigated variations in voter turnout across states and election years. Examined winning margins to assess the competitiveness of constituencies.

  - **Predictive**

  Historical voter turnout trends can help estimate future electoral participation levels. Consistently high-performing parties and candidates may continue to maintain strong electoral positions in future elections. Constituencies with narrow victory margins may experience highly competitive contests in upcoming elections.

  - **Prescriptive**

  Election authorities can focus voter awareness initiatives in regions with historically low turnout. Political parties can optimize campaign strategies based on constituency-level performance and voting patterns. Constituencies with close electoral contests may require targeted outreach and engagement strategies.

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10. Conclusions

The integration of Excel and Power BI enabled efficient processing, analysis, and visualization of Lok Sabha General Elections data. By leveraging data preparation techniques, DAX measures, interactive dashboards, and advanced visualizations, the project successfully transformed raw electoral data into meaningful insights related to voter turnout, party performance, candidate rankings, and constituency-level election outcomes. The resulting dashboard serves as an effective decision-support and election analytics tool.